Yang-Jun Joo, Ph.D.

Data Scientist | Quantitative Risk Assessment

Autonomous Vehicle Safety | Statistical & Probabilistic Modeling | Large-Scale Driving Data

Orlando, FL | Open to relocation to Foster City, CA

yangjunjoo.phd@gmail.com | +1 (813) 455-1392 | **LinkedIn** | **GitHub** | **Google Scholar**

Professional Summary: Quantitative risk assessment data scientist with 8+ years of research and applied project experience, including 3+ years of post-Ph.D. experience, developing probabilistic safety metrics, uncertainty-aware models, and automated risk assessment tools for autonomous vehicle systems. Author/co-author of 15 peer-reviewed papers and PI on two competitively funded safety projects, with hands-on experience in Python, SQL, multi-terabyte driving data, and human-behavior research.

TECHNICAL SKILLS

Programming & Data: Python, SQL, R, pandas, NumPy, scikit-learn, PyTorch, Git, Linux, ETL and analytics pipelines.

Risk & Statistics: Probabilistic modeling, Bayesian networks, reliability analysis, surrogate safety measures, regression, treatment-effect analysis, uncertainty quantification, conformal prediction.

Safety & Mobility: Risk-assessment frameworks, safety-performance metrics, AV safety evaluation, naturalistic driving video, connected-vehicle and trajectory data, traffic-conflict analysis, human factors.

PROFESSIONAL EXPERIENCE

Postdoctoral Scholar | **University of Central Florida** | Mar 2024 – Present

- Led FDOT quantitative safety workstreams with cross-functional agency and technical partners, defining and validating safety-performance metrics using high-resolution traffic-controller, connected-vehicle, and naturalistic driving data.
- Built Python and SQL pipelines for multi-terabyte datasets, analyzing high-resolution signal-controller logs from 1,000+ intersections and trajectory data across four freeways to automate and standardize risk-metric generation and reporting.
- Developed uncertainty-aware evaluation methods for autonomous-driving vision-language models using conformal prediction, identifying context-specific calibration failures in safety-critical driving-scene question answering.
- Designed and analyzed a study of 926 participants and 1,325 clips, using regression and treatment-effect analysis to quantify automation effects on passenger risk perception.

Postdoctoral Researcher | **Seoul National University** | Mar 2023 – Feb 2024

- Led research on real-time infrastructure-to-vehicle (V2I) safety information systems for the Korean National Police Agency, defining safety-critical scenarios and operational safety criteria.

Graduate Research Assistant | **Seoul National University** | Mar 2018 – Feb 2023

- Developed generalized quantitative risk assessment methods for autonomous vehicle lane changes and collision avoidance using field theory, Bayesian networks, reliability analysis, and surrogate safety measures.
- Secured and led a KRW 96.1M Global Ph.D. Fellowship, producing three first-author peer-reviewed publications.

SELECTED PUBLICATIONS

- Joo et al. (2023), "A Generalized Driving Risk Assessment Method for Autonomous Vehicles Using Field Theory," *Analytic Methods in Accident Research*, 37, 100251.
- Joo et al. (2022), "A Data-Driven Bayesian Network for Probabilistic Crash Risk Assessment of Individual Driver with Traffic Violation and Crash Records," *Accident Analysis & Prevention*, 174, 106721.
- Joo et al. (2026), "Context-Aware Conformal Prediction for VLM-Based Driving Scene QA," *Forty-Third International Conference on Machine Learning (ICML) Workshop EMM-QA*.

EDUCATION

Ph.D., Civil & Environmental Engineering (Transportation), Seoul National University, 2023

B.S., Civil & Environmental Engineering, Seoul National University, 2018